Supplementary Information for: Does a surplus-production ladder improve forecasts of Northern cod? A scored retention test against persistence on NAFO 2J3KL

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Abstract

This Supplementary Information records the material that supports the main text without carrying its argument: the pass order and protocol status (S1); the origin sets (S2); the full Diebold–Mariano and paired block-bootstrap grids with block-length sensitivity (S3); the period decomposition at five years, on the annual-landings pass, and the post-1995 pooled margins with intervals (S4); the fitted parameters of every fixed window and every rolling origin (S5); the census of floor forecasts (S6); the secondary scores (S7); the simulation design, the power decomposition, the chance baseline, the alternative rules and the failed pre-check diagnostics (S8); the decomposition of the lagged-start module's deficit (S9); the Edwards Aquifer restatement of the rule (S10); the vintage and threshold comparisons (S11); the 2022–2023 observation-suspension chronology (S12); the reproducibility and verification record (S13); and the catch-timing sensitivity (S14). Section, table and figure numbers without the prefix "S" refer to the main text.

S1. Pass order and protocol status

The study is a fixed computational protocol with a recorded execution order, not a prospective registration: no dated pre-scoring protocol file exists for the empirical passes. The order below is established from the analysis code and its archived metadata. Passes 1 and 2 are labelled in the driver

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script `run_ladder.py` and distinguished in `results/meta.json` by the keys `catch_pass1` and `catch_pass2`; passes 3 and 4 are separate scripts that import from that driver, which fixes their order after it.

Table S1: Order in which the passes were run.

#	Pass	Data treatment	Script / archived output
1	Coarse-regime catch	240/120/5 kt (through 1991 / 1992 / from 1993)	`run_ladder.py` (pass 1); `fixed_window_scores.csv`, `rolling_summary.csv` (rows `catch = regime`), `rolling_forecasts_regime.csv`
2	Annual landings	Schijns et al. (2021) Table 1, t/1000	`run_ladder.py` (pass 2); same files, rows `catch = annual`; `rolling_forecasts.csv`
2b	Survey start	Forecast initiated at the survey-scaled state, annual catch	`run_survey_start()` within pass 2; `survey_start_forecasts.csv`
3	Capelin variants	Two-regime r at the 1991 break; acoustic-index multiplier on observed years	`run_capelin_regime.py`, `run_capelin_index.py`; `capelin_*_summary.csv`
4	Specification B	The ladder on the xteNCAM series alone, not pooled	`run_xte.py`; `xte_rolling_summary.csv`, `xte_fixed_window_scores.csv`
5	Uncertainty layer	DM statistics and block bootstrap on the archived per-origin files	`campaign_e1_dm_uncertainty_regime.py`; `e1_dm_uncertainty_regime.csv`
6	Simulation	D1–D5 per `SPECIFICATION_v4.md` (locked 2026-09-10 before the code was written); D6–D7 per Amendment 1 (locked 2026-09-11)	`tools/sim_retention_power.py`; `sim_retention_power.csv`, `sim_misspecified_D6D7.csv`
7	This revision	Re-execution of passes 1–6 in a third environment; paired bootstrap confirmation; period decomposition; rolling parameter record; catch-timing sensitivity	`paperE1_cod_forecast_ladder_v69_src/work/` in the repository

Three points bear on how the scores should be read. The five ladder mod-

ules were fixed before any pass was scored: the `SPECS` list in `run_ladder.py` defines M1, M1b, M2, M3 and M4 once, Specification B imports that list rather than redefining it, and none was altered after a score was seen. The scoring core, the fixed windows, the rolling-origin RMSE comparison and the persistence bar at both horizons, was coded before the first pass. Two elements of the retention rule were formalised after the pass-1 scores had been computed: the 5% tie band and the comparator declarations for the non-nested modules (M2 against M1, M4 against M3). Neither decides any retention outcome, the smallest pooled deficit against persistence being 17.1%. The capelin pass is the one place where a model was added after scoring began (the index multiplier), and it is reported as a variant, not as a member of the ladder. Specification B was run last and never pooled with Specification A; the constraint is recorded in the code's docstring and in `meta.json`.

S2. Origin sets and windows

Table S2: Origin sets. All baselines are evaluated on the module's own origin set at each horizon. The mixed-origin persistence values that the archived Specification B summary also carries (87.6 and 317.7 kt on 63 and 59 origins) are not comparators and are not used.

Pass	Origins	n ($h = 1$)	n ($h = 5$)
Specification A, ladder (all catch passes, survey start, two-regime r)	1990–2014 ($h = 1$); 1990–2010 ($h = 5$); minimum eight training years	25	21
Specification A, capelin-index multiplier	1991–2014; 1991–2010 (origins with a carried acoustic value)	24	20
Specification B, ladder and two-regime r	1965–2023; 1965–2019; minimum twelve training years	59	55
Specification B, capelin-index multiplier	1988–2023; 1988–2019	36	32

The capelin acoustic index is observed in 1982 and 1985–2023 with gaps (1983, 1984, 1993–1995, 1997, 1998, 2006, 2016, 2020–2022 missing; 2023 = 331.3 kt). The multiplier pass carries the last observed value across gaps and drops origins with no carried value; the two-regime pass uses the 1991 capelin collapse as a declared break and allows r to differ before and after

it, with the post-break value estimated only once post-break transitions are in the training window.

S3. Uncertainty layer: full grids

Table S3 gives every module–benchmark pair of the archived uncertainty layer: the DM statistic (loss differences of squared errors; autocorrelation-consistent variance truncated at lag $h-1$; reported as a z-score), the paired moving-block bootstrap interval of the RMSE gap (block $\max(h, 3)$, 20 000 replicates, seed 0) and its two-sided p-value. The bootstrap in the archived script draws block indices from a generator seeded identically for the two loss series, so the resample is paired; re-running it with an explicitly shared index draw reproduces every interval to the printed digit. Table S4 gives the block-length sensitivity (5 000 replicates per cell) for the M1 and M3 persistence margins.

The DM z-statistic is less stable than the bootstrap under the truncation lag: for the M1 persistence margin at $h = 1$ it moves from 1.14 at lag 0 to 2.30 at lag 3 on Specification A and from 2.81 to 1.70 on Specification B, crossing the conventional 1.96 in opposite directions, whereas the bootstrap p moves within 0.17–0.19 (A) and 0.036–0.049 (B) over blocks 3–9. Four of the 28 persistence and comparator rows have a bootstrap interval excluding zero with $|z| < 1.96$ (A, $h = 1$, M4 vs M3; B, $h = 1$, M3 vs persistence; B, $h = 5$, M1b vs persistence; B, $h = 5$, M4 vs M3). No verdict rests on either statistic where they disagree; the retention rule is applied to point estimates.

Table S3: Archived uncertainty layer: all pairs. RMSE in kt; gap = RMSE(A)−RMSE(B); CI is the 95% paired block-bootstrap interval of the gap; p is the two-sided bootstrap p-value. The M2-against-M1b rows are an alternative-comparator disclosure, not part of the rule.

Spec.	h	A	B	n	RMSE(A)	RMSE(B)	gap	DM z	95% CI	p
A	1	M1	persistence	25	120.5	98.0	+22.5	1.13	[-13.4, +52.2]	0.19
A	1	M1b	persistence	25	114.8	98.0	+16.8	0.98	[-19.4, +59.6]	0.38
A	1	M2	persistence	25	144.1	98.0	+46.0	1.29	[-12.3, +83.0]	0.45
A	1	M3	persistence	25	134.6	98.0	+36.6	0.99	[-16.8, +64.2]	0.93
A	1	M4	persistence	25	195.6	98.0	+97.5	1.53	[-9.1, +172.7]	0.28
A	1	M2	M1	25	144.1	120.5	+23.5	0.91	[-60.7, +66.9]	0.68
A	1	M4	M3	25	195.6	134.6	+60.9	1.21	[+4.4, +134.4]	< 0.005
A	1	M2	M1b	25	144.1	114.8	+29.3	0.91	[-67.6, +82.8]	0.70
A	5	M1	persistence	21	288.7	264.7	+24.0	1.84	[-43.1, +56.4]	0.28
A	5	M1b	persistence	21	288.6	264.7	+23.9	1.84	[-43.1, +56.4]	0.28
A	5	M2	persistence	21	398.2	264.7	+133.5	1.10	[-21.0, +217.5]	0.42
A	5	M3	persistence	21	366.4	264.7	+101.6	1.12	[-0.8, +151.4]	0.08
A	5	M4	persistence	21	488.2	264.7	+223.5	1.14	[-20.8, +409.4]	0.40
A	5	M2	M1	21	398.2	288.7	+109.5	0.98	[-77.1, +232.8]	0.81
A	5	M4	M3	21	488.2	366.4	+121.9	1.15	[-27.4, +263.2]	0.71
A	5	M2	M1b	21	398.2	288.6	+109.7	0.98	[-77.1, +232.8]	0.81
B	1	M1	persistence	59	119.5	84.4	+35.0	2.81	[+2.7, +70.8]	0.03
B	1	M1b	persistence	59	151.6	84.4	+67.2	3.60	[+18.3, +117.7]	0.01
B	1	M2	persistence	59	166.0	84.4	+81.6	3.22	[+36.4, +122.8]	< 0.005
B	1	M3	persistence	59	126.7	84.4	+42.3	1.85	[+1.0, +92.5]	0.04
B	1	M4	persistence	59	205.7	84.4	+121.3	3.20	[+53.7, +193.0]	< 0.005
B	1	M2	M1	59	166.0	119.5	+46.6	1.78	[-20.5, +104.5]	0.17
B	1	M4	M3	59	205.7	126.7	+79.0	3.07	[+35.6, +116.4]	< 0.005
B	1	M2	M1b	59	166.0	151.6	+14.4	0.51	[-67.5, +89.1]	0.77
B	5	M1	persistence	55	431.9	300.0	+131.9	2.06	[+33.2, +218.6]	0.01
B	5	M1b	persistence	55	445.5	300.0	+145.5	1.80	[+18.2, +250.9]	0.02
B	5	M2	persistence	55	1058.9	300.0	+758.9	2.15	[+363.6, +1038.5]	< 0.005
B	5	M3	persistence	55	930.1	300.0	+630.2	2.39	[+352.8, +845.6]	< 0.005
B	5	M4	persistence	55	1030.7	300.0	+730.8	2.35	[+407.3, +978.3]	< 0.005
B	5	M2	M1	55	1058.9	431.9	+627.0	1.97	[+195.7, +929.7]	< 0.005
B	5	M4	M3	55	1030.7	930.1	+100.6	1.88	[+20.2, +177.4]	0.01
B	5	M2	M1b	55	1058.9	445.5	+613.4	1.93	[+170.9, +945.1]	< 0.005

Table S4: Block-length sensitivity of the paired bootstrap interval (kt) for the persistence margins of M1 and M3. The archived layer uses block 3 at $h = 1$ and block 5 at $h = 5$. One classification changes with the block: the M3 one-step margin on Specification B includes zero for blocks of four or more.

Spec.	h	Module	gap	block 2	block 3	block 4	block 5	block 6	block 8
A	1	M1	+22.5	[-13, +49]	[-13, +52]	[-14, +54]	[-16, +55]	[-18, +56]	[-18, +56]
A	1	M3	+36.6	[-17, +77]	[-17, +65]	[-16, +56]	[-15, +54]	[-14, +55]	[-14, +55]
A	5	M1	+24.0	[-37, +56]	[-37, +56]	[-41, +58]	[-43, +56]	[-51, +55]	[-65, +56]
A	5	M3	+101.6	[-17, +181]	[-10, +179]	[-0.4, +167]	[-0.8, +151]	[-3, +158]	[-3, +152]
B	1	M1	+35.0	[+5, +67]	[+2, +71]	[+1, +74]	[+1, +74]	[+0.1, +75]	[+1, +73]
B	1	M3	+42.3	[+4, +87]	[+1, +95]	[-1, +95]	[-3, +93]	[-4, +90]	[-4, +81]
B	5	M1	+131.9	[+62, +198]	[+49, +208]	[+37, +215]	[+33, +221]	[+25, +224]	[+13, +232]
B	5	M3	+630.2	[+436, +803]	[+394, +830]	[+362, +848]	[+348, +843]	[+310, +850]	[+272, +849]

S4. Period decomposition: five-year horizon, annual pass, pooled margins

Table S5: Five-year rolling-origin RMSE (kt) by period of the forecast origin, with origin-matched persistence; superscripts are the number of origins at which the forecast sits at the numerical floor. Companion to Table 6.

Spec.	Origins	n	persist	M1	M1b	M2	M3	M4
A	1990–1995	6	479.4	501.1^{4}	500.8^{4}	740.0^{1}	674.5^{3}	909.2^{1}
A	1996–2012	15	78.6	127.6^{15}	127.6^{15}	54.7	77.1	54.9
A	1996–2014	15	78.6	127.6^{15}	127.6^{15}	54.7	77.1	54.9
B	1965–1990	26	398.8	573.4^{10}	595.2^{17}	1186.2^{7}	1178.6^{7}	1252.0^{7}
B	1991–1995	5	301.9	211.2^{4}	27.5^{5}	1490.1	834.3^{4}	1223.5^{3}
B	1996–2012	17	123.3	178.5^{17}	178.5^{17}	250.9	217.3	217.8
B	2013–2023	7	119.9	367.8^{5}	406.6^{7}	1358.2	1012.9	1157.9
B	1996–2023	24	122.3	249.0^{22}	266.1^{24}	763.3	576.8	651.7

Table S6: Specification A, annual-landings pass: rolling-origin RMSE (kt) by period at $h = 1$ (upper block) and $h = 5$ (lower block). The low-biomass band gives the same ordering as the coarse-regime pass.

Spec.	Origins	n	persist	M1	M1b	M2	M3	M4
A	1990–1995	6	193.6	209.6^{3}	187.1^{4}	325.6	312.1^{3}	419.1
A	1996–2012	17	19.6	63.0^{12}	69.9^{13}	17.2	15.0	19.8
A	2013–2014	2	66.7	127.2	134.7	33.5	28.7	40.2
A	1996–2014	19	28.5	72.5^{12}	79.2^{13}	19.6	17.0	22.9
A	1990–1995	6	479.4	501.1^{4}	500.8^{4}	732.3^{1}	647.5^{3}	906.6^{1}
A	1996–2012	15	78.6	127.6^{15}	127.6^{15}	50.9	73.0	49.7
A	1996–2014	15	78.6	127.6^{15}	127.6^{15}	50.9	73.0	49.7

Band-level intervals for the collapse bands ($n = 5$–6) are degenerate under a block bootstrap and are not reported. A leave-one-origin-out check on the Specification B collapse band confirms that the orderings there do not depend on a single origin: M1 and M1b are below persistence in all five leave-one-out sets and M3 in four of five, but these are the floor-forecast wins discussed in Section 4.3.

Table S7: Post-1995 pooled margins against persistence (origins 1996 onward), with paired block-bootstrap intervals (block $\max(h, 3)$, 20 000 replicates). Specification A: 1996–2014 ($h = 1$) and 1996–2010 ($h = 5$); Specification B: 1996–2023 and 1996–2019. The bands were chosen after the data were seen; this table describes, it does not test.

Spec.	h	n	Module	RMSE	persist	gap	rel.	95% CI (kt)	p
A	1	19	M1	73.1	28.5	+44.6	+156%	[+28.9, +60.3]	< 0.005
A	1	19	M1b	79.8	28.5	+51.3	+180%	[+28.9, +69.5]	< 0.005
A	1	19	M2	19.4	28.5	-9.1	-32%	[-17.1, +6.6]	0.47
A	1	19	M3	16.7	28.5	-11.8	-41%	[-20.5, +3.3]	0.19
A	1	19	M4	22.1	28.5	-6.4	-22%	[-14.1, +9.0]	0.72
A	5	15	M1	127.6	78.6	+49.0	+62%	[+15.4, +68.1]	< 0.005
A	5	15	M1b	127.6	78.6	+49.0	+62%	[+15.4, +68.1]	< 0.005
A	5	15	M2	54.7	78.6	-23.8	-30%	[-27.5, +11.9]	0.76
A	5	15	M3	77.1	78.6	-1.5	-2%	[-2.7, +31.7]	0.19
A	5	15	M4	54.9	78.6	-23.6	-30%	[-27.2, +12.1]	0.78
B	1	28	M1	91.5	41.5	+50.0	+121%	[+26.3, +78.8]	< 0.005
B	1	28	M1b	124.0	41.5	+82.5	+199%	[+53.2, +117.7]	< 0.005
B	1	28	M2	102.7	41.5	+61.2	+148%	[-2.5, +93.3]	0.06
B	1	28	M3	62.3	41.5	+20.9	+50%	[-11.1, +47.0]	0.25
B	1	28	M4	87.4	41.5	+45.9	+111%	[-4.9, +73.3]	0.08
B	5	24	M1	249.0	122.3	+126.7	+104%	[+27.3, +196.8]	< 0.005
B	5	24	M1b	266.1	122.3	+143.8	+118%	[+27.3, +227.5]	< 0.005
B	5	24	M2	763.3	122.3	+641.0	+524%	[+97.9, +972.2]	< 0.005
B	5	24	M3	576.8	122.3	+454.5	+372%	[+46.6, +708.8]	< 0.005
B	5	24	M4	651.7	122.3	+529.4	+433%	[+61.7, +813.0]	< 0.005

S5. Fitted parameters

The K estimate of the coarse-regime recovery fit, 500.0, is the initialiser $K_0 = \max(1.5 \max S, 500)$: the objective is flat in K above the observed range and the single L-BFGS-B start does not move it. Under annual landings the same fit runs to the 5000-kt upper bound. In both cases the production curve over the observed range (10–81 kt) is effectively linear with slope r (Fig. 5).

Remark S1 (The compensatory branch is not the $\mathfrak{s} \to 0$ limit of the depensatory one). Substituting $\mathfrak{s} = 0$ in $a(S) = (S - \mathfrak{s})/(K - \mathfrak{s})$ gives $a(S) = S/K$, so $g(S) = rS(1 - S/K)(S/K)$, a cubic surplus, not the Schaefer form $a \equiv 1$. M1b at $\mathfrak{s} = 0$ is therefore a different model from M1, which is why the two differ slightly in the collapse window (693.6 against 694.3 kt) even though both thresholds are at zero; what M1b adds to M1 is a branch of the family, not a limiting case of it.

Table S8: Fixed-window fits (one-step least squares on the training window). ϕ is the lag-one coefficient of the training residuals, clipped to $[-0.95, 0.95]$; σ their standard deviation (kt). M2–M4 share the stock-flow fit; M1 uses the training-mean catch, M1b the depensatory branch. Bounds: $r \in [10^{-3}, 2]$, $K \in [\max S + 10,\ 5000]$, $\mathfrak{s} \in [0, \max S]$; initialiser $r_0 = 0.3$, $K_0 = \max(1.5 \max S, 500)$.

Window (catch)	Module	r	K (kt)	$\mathfrak{s}$ (kt)	ϕ	σ
A collapse, train 1983–90 (coarse)	M1	1.935	1032.7	–	−0.14	33.8
	M1b	2.000	1058.9	0.0	−0.65	40.2
	M2–M4	1.935	1032.7	–	−0.14	33.8
A recovery, train 1995–2007 (coarse)	M1	0.458	500.0	–	0.95	11.8
	M1b	2.000	105.9	0.0	0.95	11.7
	M2–M4	0.458	500.0	–	0.95	11.8
A collapse (annual landings)	M1	1.942	1034.3	–	−0.14	33.8
	M1b	2.000	1062.0	0.0	−0.66	40.5
	M2–M4	1.991	1032.3	–	−0.02	47.0
A recovery (annual landings)	M1	0.370	5000.0	–	0.95	11.8
	M1b	2.000	129.8	0.0	0.95	11.6
	M2–M4	0.390	5000.0	–	0.95	10.6

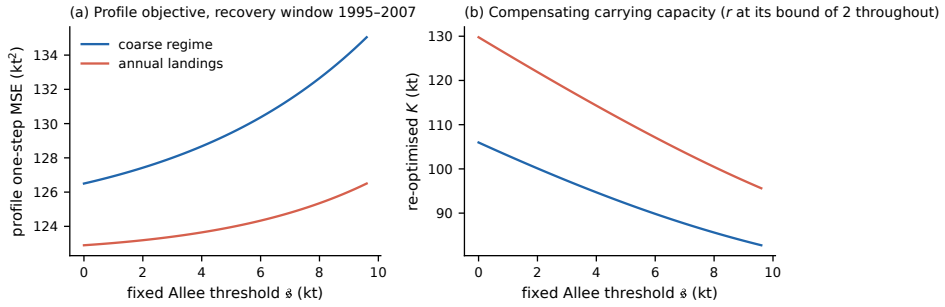


Figure S1: Profile of the one-step objective over a fixed depensation threshold $\mathfrak{s}$ in the recovery window (train 1995–2007), with r and K re-optimised at each value. (a) The objective rises monotonically from $\mathfrak{s} = 0$ and changes by 6.7% (coarse regime, 126.5 to 135.0 kt^2) and 2.9% (annual landings, 122.9 to 126.5) across the range below the smallest observed state (9.7 kt). (b) The compensating K falls as $\mathfrak{s}$ rises, with r at its bound of 2 throughout. The threshold is unidentified in this window; its least-squares value is the boundary.

Table S9: Specification A, coarse regime: parameters refitted at every rolling origin (expanding window from 1983). $\bar{C}$ is the training-mean catch used by M1 and M1b. From the 1992 origin, once the collapse transitions enter the training window, K sits at its bound for the stock-flow fit and the depensatory threshold moves to 585–734 kt; from 2011 it falls to about 5 kt.

	M2–M4 (stock-flow)			M1 (autonomous)			M1b (depensatory)		
Origin	r	K	ϕ	$\bar{C}$	r	K	r	K	s
1990	1.935	1033	-0.14	240.0	1.935	1033	2.00	1059	0.0
1991	1.557	1058	-0.03	240.0	1.557	1058	2.00	1042	0.0
1992	0.269	5000	+0.34	228.0	0.252	5000	2.00	1192	672.7
1993	0.250	5000	+0.67	207.7	0.212	5000	2.00	2110	635.0
1994	0.248	5000	+0.57	190.8	0.190	5000	2.00	3049	585.3
1995	0.248	5000	+0.56	176.5	0.170	5000	2.00	3300	597.9
1996	0.248	5000	+0.56	164.3	0.153	5000	2.00	3271	628.8
1997	0.248	5000	+0.56	153.7	0.138	5000	2.00	3332	649.2
1998	0.248	5000	+0.56	144.4	0.125	5000	2.00	3453	663.6
1999	0.248	5000	+0.56	136.2	0.114	5000	2.00	3633	673.3
2000	0.248	5000	+0.56	128.9	0.104	5000	2.00	3942	676.1
2001	0.248	5000	+0.56	122.4	0.095	5000	2.00	4198	681.7
2002	0.304	2537	+0.58	116.5	0.087	5000	2.00	4373	690.6
2003	0.247	5000	+0.56	111.2	0.080	5000	2.00	4476	701.6
2004	0.247	5000	+0.56	106.4	0.073	5000	2.00	4547	713.0
2005	0.247	5000	+0.56	102.0	0.067	5000	2.00	4606	723.7
2006	0.248	5000	+0.56	97.9	0.061	5000	2.00	4768	730.7
2007	0.248	5000	+0.56	94.2	0.057	5000	2.00	5000	734.1
2008	0.248	5000	+0.56	90.8	0.054	5000	1.57	5000	707.8
2009	0.248	5000	+0.56	87.6	0.050	5000	1.12	5000	656.9
2010	0.247	5000	+0.56	84.6	0.047	5000	0.80	5000	588.7
2011	0.246	5000	+0.56	81.9	0.044	5000	0.13	1411	5.3
2012	0.247	5000	+0.56	79.3	0.042	5000	0.12	1411	5.3
2013	0.247	5000	+0.56	76.9	0.083	1408	0.11	1411	5.2
2014	0.249	5000	+0.56	74.7	0.148	1114	0.11	1411	5.1

Table S10: Specification B: range of the refitted parameters over the origins of each band. The depensatory threshold stays at or near zero throughout; the stock-flow K is at or near its bound after 1993.

		M2–M4		M1			M1b		
Origins	n	r	K	$\bar{C}$	r	K	r	K	s
1965–1990	26	0.46–0.78	2482–5000	354–468	0.77–0.95	1845–2276	1.41–2.00	1642–2279	0.00–0.27
1991–1995	5	0.50–0.59	3355–4800	317–349	0.71–0.83	1869–1922	1.57–1.76	1654–1660	0.00–0.00
1996–2012	17	0.50–0.50	4802–5000	227–310	0.56–0.70	1789–1905	0.91–1.53	1651–2261	0.00–0.04
2013–2023	11	0.47–0.50	5000–5000	193–223	0.53–0.57	1682–1774	0.81–1.09	1603–2261	0.00–0.04

S6. Floor-forecast census

A forecast is at the floor when the iterated map reaches the numerical floor $\epsilon = 10^{-3}$ kt; its absolute error then equals the observed SSB. Table S11 counts floor forecasts by band from the archived per-origin files (coarse regime for Specification A). The counts in the first block are those behind the superscripts of Table 6 and Table S5.

Table S11: Number of origins at which the forecast is at the floor, by band and horizon.

Spec.	Origins	n	M1	M1b	M2	M3	M4
A, $h = 1$	1990–1995	6	3	4	0	3	0
A, $h = 1$	1996–2012	17	12	13	0	0	0
A, $h = 1$	2013–2014	2	0	0	0	0	0
A, $h = 5$	1990–1995	6	4	4	1	3	1
A, $h = 5$	1996–2010	15	15	15	0	0	0
B, $h = 1$	1965–1990	26	2	3	0	0	0
B, $h = 1$	1991–1995	5	3	3	0	3	1
B, $h = 1$	1996–2012	17	17	17	0	0	0
B, $h = 1$	2013–2023	11	0	1	0	0	0
B, $h = 5$	1965–1990	26	10	17	7	7	7
B, $h = 5$	1991–1995	5	4	5	0	4	3
B, $h = 5$	1996–2012	17	17	17	0	0	0
B, $h = 5$	2013–2019	7	5	7	0	0	0

On Specification A the one-step M1 forecast is at the floor at every origin from 1993 to 2007; at 2008–2012 it forecasts 13–37 kt from states of 91–169 kt, and at 2013–2014 it forecasts 105 and 204 kt against 250 and 299. The mechanism is the constant catch: at the 2000 origin the training-mean catch is 129 kt (coarse) or 122 kt (annual) against a state of 34 kt, and at the 2010 origin 85 or 79 kt against 91 kt. On Specification B the training-mean catch is 317 kt at the 1995 origin and still 201 kt at the 2020 origin.

The log-RMSE diagnostic of the archived summaries uses $\log \max(\hat{S}, \epsilon)$ and is dominated by these floor forecasts; over the whole origin sets the floor binds for M1 at 15 of 25 ($h = 1$) and 19 of 21 ($h = 5$) origins on the Specification A annual pass and at 22 of 59 and 36 of 55 on Specification B, for M1b at 17/25, 19/21, 24/59 and 46/55, and for M2–M4 at between zero and eleven origins per cell. The raw RMSE, not the log column, is the retention score.

Table S12: Brier score of the hard below-threshold indicator implied by the point forecast ($\hat{S} <$ threshold against $S <$ threshold), origin-matched. Specification A uses the LRP of 884.6 kt, below which every origin and target after 1990 lies, so the indicator never varies and persistence scores zero trivially; Specification B uses $B_{\mathrm{lim}} = 276$ kt, which the series crosses in 1977, 1980, 1993 and 2016.

	A, $h = 1$	A, $h = 5$	B, $h = 1$	B, $h = 5$
persistence	0.000	0.000	0.068	0.291
M1	0.040	0.048	0.051	0.455
M1b	0.000	0.048	0.153	0.473
M2	0.080	0.143	0.068	0.364
M3	0.080	0.095	0.017	0.309
M4	0.080	0.143	0.068	0.382

S7. Secondary scores

On Specification B the M3 indicator has the lowest one-step Brier score (0.017 against 0.068 for persistence: one misclassified crossing instead of four), and M1's five-year score (0.455) is the worst because it forecasts the floor through two decades in which the stock sat just below B_{lim} and then crossed it. The direction-of-change score, the fraction of consecutive targets whose observed and forecast changes agree in sign, is 0.80–1.00 for the structural modules on Specification A and undefined for persistence (its forecast change is zero); it separates modules on an outcome that scarcely varies and carries no retention weight.

An interval-scale comparison is available on Specification B because the assessment publishes 95% limits for its own predictand, with median relative width 0.44. Writing $D = |\hat{S}_{t+h|t} - S_{t+h}| / [(\mathrm{hi}_{t+h} - \mathrm{lo}_{t+h})/2]$, the proportion of hindcast errors not exceeding half the published width is 56% for persistence, 36% for M1 and 59% for M3 at $h = 1$ ($n = 59$), and 18%, 5% and 7% at $h = 5$ ($n = 55$). These are not coverage probabilities: the published interval expresses uncertainty about true biomass given the assessment's data, re-centring its width on a hindcast gives an interval with no established coverage, and the limits are right-skewed on the biomass scale (67 of 71 years asymmetric by more than 10%). The statistic indicates only that hindcast discrepancies are of the same order as, or larger than, reported target-state uncertainty, and that M3 reads marginally above persistence on it at one step where the RMSE ordering does not.

Table S13: Data-generating processes. All series have $T = 33$ years (the Specification A length) with Gaussian process noise $\sigma \in \{11.8, 33.8\}$ kt entering the state update; parameters are the archived fixed-window fits. Rolling origins use the eight-year minimum training length of Specification A. M3 and M4 were not simulated as truths.

DGP	Generating process	Source of parameters
D1	M1 autonomous Schaefer, $r = 1.935$, $K = 1032.7$, $C \equiv 240$, $S_0 = 900$	collapse-window coarse fit
D2	M1 autonomous Schaefer, $r = 0.458$, $K = 500$, $C \equiv 5$, $S_0 = 30$	recovery-window coarse fit
D3	M2 stock-flow, $r = 1.935$, $K = 1032.7$, catch 240/120/5 along the Specification A schedule, $S_0 = 900$	collapse fit and regime path
D4	M1b depensatory, $r = 0.458$, $K = 500$, $\mathfrak{s} = 15$, $C \equiv 5$, $S_0 = 30$	recovery fit with an identifiable threshold
D5	Persistence-true null, $S_{t+1} = S_t + \eta_t$, no production, $C \equiv 0$, $S_0 = 300$	–
D6	Drifting productivity, $r_t = 1.935\, e^{-0.05t} + 0.35$, $K = 1032.7$, catch 55% of the instantaneous $r_t K/4$	Amendment 1 (outside the class)
D7	Smooth path: noise-free state at $r = 0.9$, $K = 1032.7$, $C = 180$, scored series = state plus Gaussian noise	Amendment 1 (outside the class)

S8. Simulation: design and full decomposition

The design sheet (`SPECIFICATION_v4.md`) was locked at commit `d5d9758` on 2026-09-10 and the simulation code was written afterwards to implement it; no design element was changed after execution. Amendment 1 (D6, D7, and a $T = 71$ series-length cell for D1 and D5) was locked at commit `bd0008f` on 2026-09-11; the $T = 71$ cell was started and abandoned as infeasible (45.6 s per rolling pass against 7 s at $T = 33$), so no series-length sensitivity is claimed. A module counts as retained in a replicate if it passes (H1)–(H3) at both horizons exactly as in the empirical rule.

In absolute terms the dominant failure is the persistence hurdle: a true module often does not out-predict persistence on data it generated. Conditional on clearing it, the comparator hurdle removes most of the survivors for the stock-flow and depensatory truths. On D3 the stock-flow module's

Table S14: Where the power is lost (averaged over the two noise levels). "H2 alone" is the persistence hurdle with the band at both horizons, ignoring the comparator.

DGP	truth	passes H2 alone	passes full rule	removed by comparator	conditional on H2
D1	M1	0.975	0.975	0.000	0%
D2	M1	0.420	0.420	0.000	0%
D3	M2	0.328	0.100	0.228	69%
D4	M1b	0.180	0.010	0.170	94%

Table S15: Chance baseline: proportion of replicates in which the generating module has the lowest one-step RMSE among the five modules (chance = 0.20).

DGP	true module ranks first	against chance
D1 (M1 true)	0.627	far better
D2 (M1 true)	0.645	far better
D3 (M2 true)	0.038	worse than chance
D4 (M1b true)	0.258	barely better

mean one-step error (85.7 kt) exceeds that of the residual (77.9) and depensatory (79.9) modules, so estimation noise actively disadvantages the correct structure relative to its siblings at this sample size. Under the four structural processes the false-retention rate is 0.044 per module–replicate pair (0.176 falsely retained modules per replicate); under the persistence-true null it is 0.005 per module. Under D6 and D7 the module falsely retained is overwhelmingly M1 (134 and 151 of 200 replicates under D6, 195 and 185 under D7): a constant-productivity map out-predicts persistence on a smoothly drifting or smoothly observed series often enough to clear both hurdles. The alternative decision rules of Table 8 were evaluated on the same archived replicates without refitting.

Two candidate diagnostics for predicting from the data alone whether the rule has power were examined and both fail. The dispersion of scores across modules (median coefficient of variation) correlates with true-module power at Spearman $\rho = 0.52$, too weakly to license a protocol. The margin of the best-scoring module over persistence correlates at $\rho = 0.88$ (Pearson $r = 0.94$) but is computed from the same out-of-sample scores the rule consumes, so it cannot gate the decision, and thresholding it at the tie band misclassifies both D3 cells, flagging "apply" where the generating module wins less often than chance. Identifying when the rule has power remains open.

S9. Decomposition of the lagged-start deficit

M4 differs from M3 in two ways: it starts the path from S_{t-1}, and it therefore integrates one more catch year. A delayed-persistence control, $\hat{S}_{t+h} = S_{t-1}$, isolates the information delay. Its one-step RMSE is 184.4 kt on Specification A and 158.0 kt on Specification B, against 98.0 and 84.4 for ordinary persistence; the delay alone therefore costs 86.4 of M4's 97.5-kt deficit against persistence on A and 73.6 of 121.3 on B. The lagged start is an information-delay penalty, not a dynamical one, and M4 is not retained for the same reason that a one-year-stale persistence forecast would not be.

S10. The rule applied elsewhere: Edwards Aquifer restatement

The same retention rule, with the same hurdles, band and horizons, was applied in Abaee (2026, Does a one-pool water-balance model improve forecasts of Edwards Aquifer head) to a one-pool water-balance forecaster of the J-17 index well. There the module's one-step RMSE was 12.84 ft against 13.23 ft for persistence, a 2.96% improvement inside the 5% band, and at five years the ordering reversed (21.25 against 21.11 ft). Retention was withheld by the band and the two-horizon requirement. On the cod series no module comes within the band at either horizon, so the ranking alone decides. The two applications share the rule and nothing else; neither series is pooled with the other and no verdict is transferred.

S11. Vintages and thresholds

Vintage overlap, 1983–2015. The NCAM Table A2 vintage (Specification A) and the RAM Legacy v4.66 continuation of the same assessment family (stock COD2J3KL) differ over 1983–2015 by +24.7 kt on average (Table A2 higher), by 26.8 kt in mean absolute value and by 131.2 kt at most, in 1984 (Fig. S2). The xteNCAM revision differs from Table A2 over the same years by 126.2 kt RMSE, with Table A2 higher by 58.5 kt on average. The 2015 origin is 298.65 kt on Table A2, 277 kt on RAM and 273 kt on xteNCAM.

Vintage-shifted LRP. The LRP defined as the 1983–1989 mean is 884.6 kt on the Table A2 vintage and 790.0 kt on the RAM vintage, a shift of 94.6 kt in the reference point from a revision of the same model family. Persistence from the 2015 origin under-predicts every outcome year 2016–2021 in both vintages: MAE 129.2 kt and RMSE 133.3 kt from the RAM origin of 277 kt, MAE 107.5 kt from the Table A2 origin. The outcome years agree across the

RAM and xteNCAM vintages to within 1–26 kt (mean absolute difference 16.8 kt).

Threshold classification on one file. The xteNCAM file (1954–2024, 71 years) carries a ratio column that fixes B_{lim} at 276 kt (per-year implied values 250–289, mean 275.5). Classifying each year under $B_{\text{lim}} = 276$ kt and under the 884.6-kt LRP gives 45 years at or above B_{lim}, 16 years at or above the LRP (1954–1969 only), and 29 disagreement years in two blocks: 1970–1992 without 1977–1979, and every outcome year 2016–2024. Under the 1980s-mean threshold the stock stands below the reference point for 55 consecutive years (1970–2024); under B_{lim} it is below for 26 years (1977–1979 and 1993–2015) and above from 2016 onward. The late-1970s episode (ratios 0.69–0.95) is a critical-zone reading, on the current model, of years the contemporary fishery regarded as healthy. The checkpoints 2005, 2017 and 2024 read 26, 451 and 342 kt. The October 2023 framework revision, which moved the stock's reported zone from critical to cautious without a change in the stock, is this experiment executed by the assessment authority. The scored RMSE comparison never uses the thresholds; every normative verdict layered on the forecasts, the Brier indicator included, inherits the dependence.

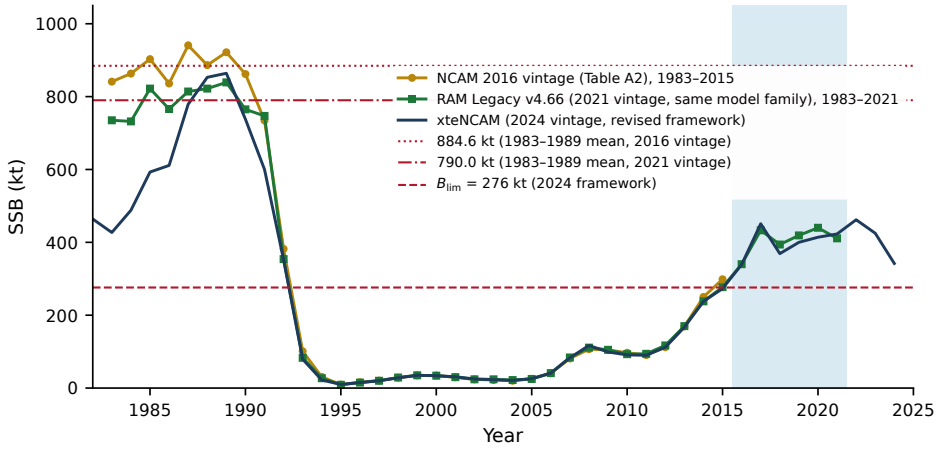


Figure S2: Three vintages of Northern cod SSB over 1983–2024: NCAM Table A2 (Specification A, 1983–2015), RAM Legacy v4.66 (the same assessment family continued to 2021) and xteNCAM Table 17 (Specification B). Horizontal lines mark the two Specification A LRP values (884.6 kt on Table A2, 790.0 kt on the RAM vintage) and $B_{\text{lim}} = 276$ kt. The outcome years 2016–2021 lie above the 2015 origin in every vintage and below 884.6 kt in every vintage.

S12. The 2022–2023 observation suspension

The test conditions on a complete annual series. The stock's real 2022–2023 history violates that condition. Partial coverage by the fall research-vessel survey in 2021 and 2022 left the leading indicator of stock size uncomputable; no typical assessment update was possible in 2022 or 2023; the stewardship maximum was rolled over unchanged at 12,999 t; and the survey vessel that carries the series was condemned in February 2023. Table S16 assembles the episode from the assessment of record and the public record.

Table S16: The 2022–2023 observation suspension and its resolution (sources: Regular et al. 2025 and its Table 17 file; the 2024 reopening announcement; the 2025 and 2026 assessment coverage).

Period	Event
2021–22	Partial fall-survey coverage; leading indicator not computable
2022–23	No typical assessment update; stewardship maximum held at 12,999 t
Feb 2023	Survey vessel condemned
Mar 2024	Revised-framework assessment: SSB at 1.2 B_{lim} (95% CI 0.7–2.1; 22% critical-zone probability)
Jun 2024	Commercial fishery reopened at 18,000 t after 32 years
2025	TAC 38,000 t; reference point revised after being deemed overestimated
Apr 2026	Healthy-zone report: SSB $\approx$ 540 kt (CI 420–700); 70% probability above the upper stock reference

Read against a review-timing rule under which a review must land no later than the worst-case loss of the information it consumes, the episode is that rule's boundary case in the wild: the information loss ran past the review cadence, and the removals rule froze for two cycles while the observation map was dark. Abaee (2026, Periodic review as sampled governance) states the formal identity; the 2022–2023 suspension is its empirical instance, and it marks the practical limit of any scoring exercise that presumes the series exists.

S13. Reproducibility and verification record

Determinism. All computations are deterministic: within a fixed interpreter and library stack, repeated executions reproduce every result file byte for byte. An independent re-execution verified all 29 recorded output checksums and regenerated all 30 result files byte-identically; one archived file carries no pinned checksum and is covered by the regeneration comparison.

Across environments (Python 3.12, numpy 2.1.3, scipy 1.14.1 against the recorded original environment), the intervention runners and the OLS-based prediction runners regenerate their outputs byte for byte, while the four optimiser-based forecast runners (L-BFGS-B fits) reproduce every scored row at printed precision with one exception: the M1b rolling row on Specification B, whose depensatory optimum is environment-sensitive at the ±17 kt level ($h = 1$: 151.6 against 153.2 kt; $h = 5$: 445.5 against 462.5 kt). That sensitivity is the identification fragility of Section S5 and is immaterial to the verdict, which persistence wins at both horizons by margins far larger than the instability.

Third environment (this revision). For this revision the four forecast runners (`run_ladder.py`, `run_xte.py`, `run_capelin_regime.py`, `run_capelin_index.py`), the uncertainty script and the simulation outputs were re-executed or re-read in a third environment (Python 3.13.16, numpy 2.3.5, scipy 1.17.1). Every result file, including the M1b rows, was reproduced at printed precision, and the persistence baselines were recomputed from the series on the archived origin sets and matched the archived origin-matched values (98.05, 264.72, 84.43, 299.98 kt). The paired block bootstrap was re-run from the archived per-origin files with an explicitly shared index draw and reproduced every interval of Table S3. The period decomposition, the rolling parameter record, the floor census and the catch-timing sensitivity were computed in the same environment from the archived files and the unmodified runner functions; their scripts are archived under `paperE1_cod_forecast_ladder_v69_src/work/`.

Verification script. `paperE1_cod_forecast_ladder_v69_verification.py` (standard library only) regenerates every number of Section 4.7 and of Sections S11–S12 from the archived data files (NCAM Table A2, RAM v4.66 time series, xteNCAM Table 17, the capelin index) and asserts the cross-checks, including the presence in the manuscript source of the pinned values and the suspension facts. All checks pass on the main text and this document taken together.

Scripts.

```
python3 wave_e_cod/src/run_ladder.py
python3 wave_e_cod/src/run_xte.py
python3 wave_e_cod/src/run_capelin_regime.py
python3 wave_e_cod/src/run_capelin_index.py
python3 "batch 7 .../campaign_e1_dm_uncertainty_regime.py"
```

```
python3 tools/sim_retention_power.py
python3 paperE1_cod_forecast_ladder_v69_src/work/paired_bootstrap.py
python3 paperE1_cod_forecast_ladder_v69_src/work/post1995.py
python3 paperE1_cod_forecast_ladder_v69_src/work/catch_timing_sensitivity.py
python3 paperE1_cod_forecast_ladder_v69_src/work/make_figures.py
```

S14. Catch-timing convention: a sensitivity outside the archived protocol

In `run_ladder.py` the estimator pairs the increment $S_{j+1} - S_j$ with the catch C_j of the same year, whereas the forecast path at origin t applies $C_{t+1}, \ldots, C_{t+h}$ to the steps $t \to t+1, \ldots, t+h-1 \to t+h$. The two conventions differ by one year. The estimation convention is the one under which SSB is a start-of-year quantity and C_t the removals during year t; the forecast convention gives the catch-supplied modules the catch of the target year, which under that reading occurs after the predictand is realised. The offset was noticed during the re-execution for this revision. Table S17 re-runs M2–M4 with the estimation convention in the forecast path (persistence and M1/M1b are unaffected). Under the coarse regime the two conventions differ only at the 1990–1992 origins, where the regime boundaries fall; under annual landings and on Specification B they differ at every origin.

Under the estimation convention every one-step deficit shrinks and no five-year deficit changes sign. M3 is 7% above persistence at one year on Specification A (coarse) and 12% on Specification B, still outside the 5% band, and 29% and 178% above at five years. The post-1995 bands of Table 6 are unchanged on Specification A (the catch is constant at 5 kt there under the coarse regime); on Specification B the post-1995 one-step M3 margin becomes +10.4 kt with interval $[-20.2, +46.8]$ ($p = 0.53$), M2 and M4 remain significantly worse ($+62.0$, $[+24.6, +98.7]$; $+45.9$, $[+11.8, +86.4]$), and all five-year margins remain far outside the band. The verdict of the main text does not depend on the convention; the size of the one-step deficits does, and a future application of the ladder should use the estimation convention throughout.

Table S17: Rolling-origin RMSE (kt) of the catch-supplied modules under the archived forecast convention (C_{t+1} applied to the step $t \to t+1$) and under the estimation convention (C_t), with the ratio to origin-matched persistence in parentheses.

Spec. (catch)	Module	h	archived	estimation convention	persistence
A annual	M2	1	160.4 (1.64)	133.3 (1.36)	98.0
A annual	M2	5	393.8 (1.49)	390.8 (1.48)	264.7
A annual	M3	1	153.6 (1.57)	125.5 (1.28)	98.0
A annual	M3	5	351.5 (1.33)	352.2 (1.33)	264.7
A annual	M4	1	206.3 (2.10)	161.9 (1.65)	98.0
A annual	M4	5	486.4 (1.84)	438.7 (1.66)	264.7
A coarse	M2	1	144.1 (1.47)	113.4 (1.16)	98.0
A coarse	M2	5	398.2 (1.50)	367.8 (1.39)	264.7
A coarse	M3	1	134.6 (1.37)	105.2 (1.07)	98.0
A coarse	M3	5	366.4 (1.38)	341.5 (1.29)	264.7
A coarse	M4	1	195.6 (1.99)	138.9 (1.42)	98.0
A coarse	M4	5	488.2 (1.84)	409.8 (1.55)	264.7
B	M2	1	166.0 (1.97)	146.4 (1.73)	84.4
B	M2	5	1058.9 (3.53)	981.8 (3.27)	300.0
B	M3	1	126.7 (1.50)	94.9 (1.12)	84.4
B	M3	5	930.1 (3.10)	832.5 (2.78)	300.0
B	M4	1	205.7 (2.44)	130.3 (1.54)	84.4
B	M4	5	1030.7 (3.44)	898.3 (2.99)	300.0

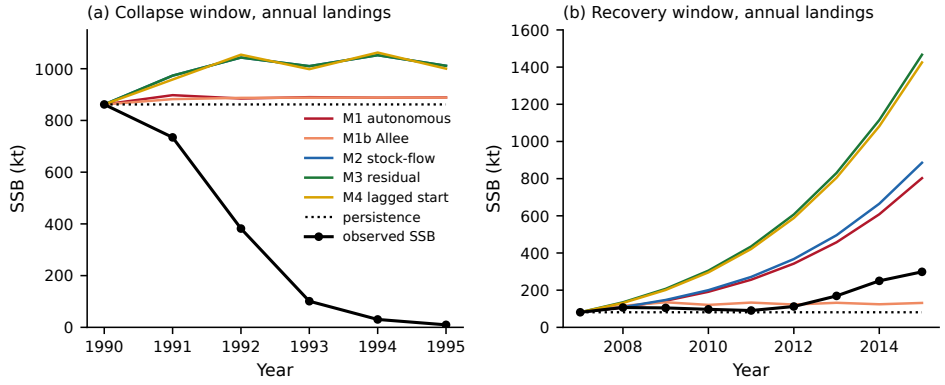


Figure S3: Fixed-window forecast paths on Specification A with annual landings (companion to Fig. 3). (a) Collapse window: as under the coarse regime, every module forecasts continuation or growth. (b) Recovery window: with K at its 5000-kt bound, production does not saturate and the catch-supplied modules over-predict more than under the coarse regime (M3 609 kt RMSE against 220); M1b is again a flat line at its fitted K.